Analog Signal Conditioning Circuit for Interference Suppression

1. Introduction

Modern industrial and autonomous systems often operate in environments containing strong electrical interference from switching electronics, RF emitters, and motor drives. These disturbances contaminate sensor outputs, making reliable signal extraction challenging.

This project focuses on designing an **analog signal conditioning circuit** capable of extracting a useful signal from a composite input containing multiple interference components and broadband noise. The design uses **active analog filters implemented with operational amplifiers** and is validated using **LTspice simulations**.

The final circuit suppresses unwanted spectral components while ensuring that the desired signal dominates the output.

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2. Objective

The objective of this project is to design and simulate an **analog signal conditioning module** that extracts a useful signal component from a composite analog input signal.

The system must:

- Accept a **standardized composite input signal**
- Suppress interference and broadband noise
- Extract and amplify the **usable signal component**
- Provide **stable and repeatable output behavior**
- Allow **manual adjustment through a control element**
- Demonstrate performance using **frequency-domain (FFT) analysis**

The design must use **pure analog components**, such as operational amplifiers, resistors, and capacitors, without using digital signal processing or programmable devices.

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3. Input Signal Characteristics

The black-box signal source produces a composite analog signal containing multiple deterministic components and broadband noise.

Component	Type	Frequency	Amplitude
Low-frequency drift	Sine	600 Hz	0.8 Vpp
Usable signal	Sine	6 kHz	0.4 Vpp
Structured interference	Square wave	18 kHz	1.2 Vpp
High-frequency interference	Sine	55 kHz	2.0 Vpp
Broadband noise	White noise	≥100 kHz	150 mVrms

The goal of the conditioning circuit is to **extract the 6 kHz signal and suppress all other components**.

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4. Design Approach

The proposed solution uses **active band-pass filtering** implemented using operational amplifiers.

The circuit consists of:

1. **Input Buffer Stage**
    - Prevents loading of the signal source
    - Maintains signal integrity
2. **First Active Filter Stage**
    - Implements a state-variable band-pass filter
    - Reduces low-frequency drift and part of the noise spectrum
3. **Second Active Filter Stage**
    - Further refines the passband
    - Attenuates high-frequency interference

By cascading two filter stages, the circuit achieves:

- Higher selectivity
- Better suppression of interference

- Improved signal-to-noise ratio

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5. Circuit Implementation

The circuit is implemented using **LT1057 operational amplifiers** with resistor-capacitor networks forming active filters.

Key Component Values

Component	Value
Capacitors	10 nF
Integrator resistors (Stage 1)	2.65 kΩ
Integrator resistors (Stage 2)	16 kΩ
Feedback resistors	10 kΩ
Input resistor	1 kΩ

The cascaded stages shape the overall frequency response to isolate the desired signal band.

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6. Simulation Setup

The circuit was simulated using **LTspice** with the provided composite input signal source.

Two types of analysis were performed:

Transient Analysis

This analysis observes the time-domain behavior of the circuit.

The input waveform contains multiple frequency components and noise, while the output waveform shows a filtered signal with reduced interference.

FFT Analysis

FFT analysis is used to evaluate frequency-domain performance.

Plots were generated for:

- Input signal spectrum
- Output signal spectrum

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7. Results

Time-Domain Results

The transient simulation shows that:

- The input signal contains large fluctuations and noise.
- The output waveform becomes a **cleaner sinusoidal signal** corresponding to the filtered frequency band.

Frequency-Domain Results

FFT plots demonstrate that:

- The input signal contains multiple frequency components.
- The output spectrum shows a **dominant peak near the target frequency region**.
- High-frequency interference and broadband noise are significantly attenuated.

This confirms the effectiveness of the signal conditioning circuit.

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8. Stability and Performance

The designed circuit demonstrates:

- Stable operation without oscillation
- No clipping or saturation under normal operating conditions
- Smooth signal behavior during operation
- Effective attenuation of unwanted frequencies

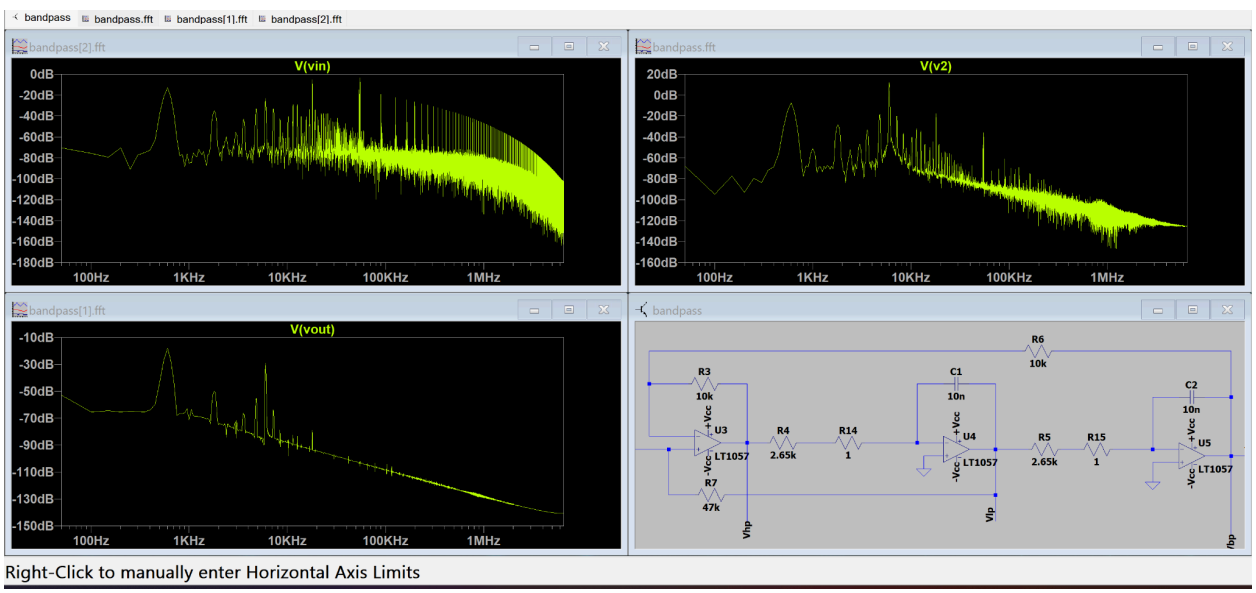
The filter stages ensure that the usable signal becomes the dominant spectral component at the output.

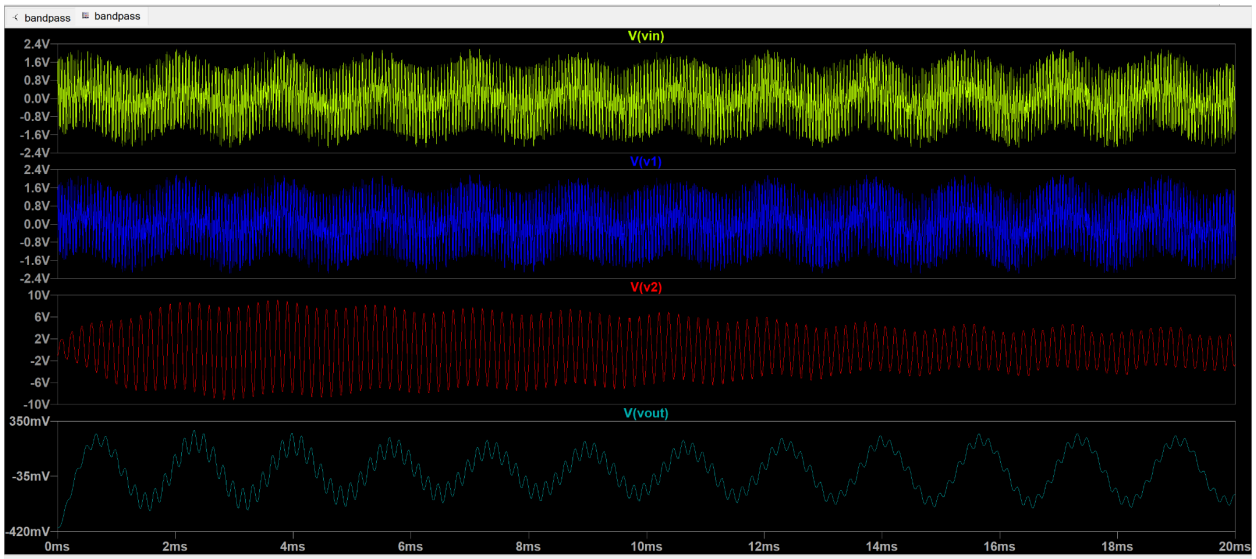
9. Conclusion

An analog signal conditioning circuit was successfully designed and simulated to extract a useful signal from a noisy composite input. By using cascaded active band-pass filters, the circuit effectively suppresses interference and noise while preserving the desired signal component.

Simulation results confirm that the circuit meets the functional requirements and demonstrates reliable analog signal extraction in the presence of multiple interference sources.

POTIOMETER defined by R4 and R5 (dual gang pots)





x = 8.42ms y = -1.058V